Assignment-3

Q1. A 1-ϕ transmission line has a resistance of 0.2 Ω and an inductive reactance of 0.4 Ω. Find the voltage at the sending end to give 500 kVA at 2 kV at receiving end at load power factors of

(i) Unity
(ii) 0.707 lagging.

Illustrate with suitable phasor diagrams.

Q2. A 3-ϕ transmission line has the following data

Resistance of each conductor = 20 Ω
Inductive reactance of each conductor = 40 Ω
Capacitive susceptance to neutral = 0.0004 ℧.
Load at the receiving end = 12,500 kVA at 0.8 lagging
Voltage at receiving end = 66 kV.

Find
(a) Sending end voltage and current
(b) Sending end power factor
(c) efficiency (d) Regulation of the line.

Use nominal-T method and draw the phasor diagram.

3. A 50 Hz, 3-ϕ transmission line of length 30 km has a total series impedance of $(40+j125)\,\Omega$ and shunt admittance of 10^{-3} mho find the sending end voltage. Use nominal π representation.

4. Determine the corona loss for 3-ϕ line operating at 110 kV which has conductors of 1.25 cm dia arranged in a 3.05 m delta configuration. Assume polished wires and air density factor of 1.07.

5. A, 3-ϕ transmission line, 160 km long has the following constants:

Resistance per phase per km = 0.2 Ω

Reactance per phase per km = 0.3127 Ω

Shunt admittance per phase per km = 1.875×10^{-6} S

Determine the sending end voltage and current by rigorous method when the line is delivering a load of 25 MVA at 0.8 power factor lagging. The receiving end voltage is kept constant at 110 kV.